

WATERWATCH · CALCULATION

WaterWatch sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (energy, solar, enclosure temperature, flushing and water use, chlorine error budget, turbidity settling, data, alert latency, wind, task times, cost)

WaterWatch sizing calculations

On paper, WaterWatch meets eight of its thirteen requirements (five by calculation, three by design), has three at risk and misses two. The misses are both about chlorine. R1 (free chlorine accuracy) is not met: with a site pH entered by hand, the uncertainty in pH alone exceeds the ± 0.1 mg/L target at pH 7.5 and above, before any sensor drift is counted. R10 (monthly maintenance) is not met on present evidence, because the chlorine sensor's drift, and so its recalibration interval, is unknown. The three at risk are turbidity (R2, fouling), outdoor survival (R9, battery temperature) and installation time (R11). The calculations changed four parts of the TRL 2 concept: the flow cell is half its former volume so that the flush clears it, a pressure-compensating regulator replaces the fixed restrictor, the settling wait before the turbidity reading rises from 10 s to 30 s, and a ventilated sun shield is added so that the battery can charge on hot days. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that water is safe to drink, and they are not a substitute for laboratory checks of the sensors, backflow tests on the sample line or electrical safety checks on the battery. See WWT-PRC-001, Safety.

■ Scope and method

The note checks every requirement in WWT-REQ-001 v0.3 against the design in WWT-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and derived dimensions, so the cell volume, tube run, enclosure size, pole and footing used here are the ones in the STEP files and in drawing WWT-DWG-001. The script also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is a public tapstand on a chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi), ambient 0 to 45 °C, hourly readings and uploads every 4 h over LTE-M or NB-IoT.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Electronics	80 mA average at 3.3 V while awake; 50 μ A sleep floor including potentiostat bias, charger and protection board; 20 mWh per upload including network attach; latching valve 2 pulses of 50 ms at 12 V, 0.5 A through an 80 % boost	Typical ESP32-S3 and SIM7000-class figures; to confirm from datasheets and measurement
Reading cycle	90 s flush with chlorine averaged over the last 10 s; 30 s settling; 5 s turbidity; 5 s for temperature and logging; 130 s awake	Section F sets the 30 s
Margin	1.5 on daily energy for alerts, retries and cold weather	As at TRL 2
Battery	LiFePO ₄ 1S2P, 3.2 V, 6 Ah; 80 % usable; 85 % of capacity at 0 °C	Typical cell data
Solar	5 W panel; 4.5 peak sun hours on a clear day, 2.5 on an overcast day; 0.6 derating for heat, dust, angle and a small charger	Screening values
Enclosure heat	Combined convection and radiation 10 W/m ² K; direct sun 900 W/m ² , diffuse 100 W/m ² ; absorptance 0.45 (light grey) or 0.70 (dusty); back face against the mounting plate; sun square to the front face	Handbook ranges; worst azimuth
Sun shield	Ventilated white hood with a 25 mm gap lets a quarter of the solar gain reach the enclosure	Assumed; to be measured
Flow	0.5 L/min pressure-compensating regulator, +10 % tolerance, compensating above about 0.7 bar; well-mixed cell (worst case for flushing)	Typical drip-irrigation flow regulators
Chlorine	HOCl pKa from Morris (1966); sensor responds to HOCl; calibrated at the site against a DPD comparator read to ± 0.07 mg/L; residual temperature error 2.5 %/K times ± 0.5 K; site pH may move ± 0.2 between visits; a pH probe variant reads to ± 0.1	pH drift and sensor coefficients are assumptions, to confirm with partner data and tests
Wind	35 m/s gust; drag coefficients 1.2 (panel, pole) and 1.3 (enclosure); medium sand, 18 kN/m ³ , $K_p = 3$	Screening values, not a code check

■ A. Daily energy (R7, R4)

- **Per reading.** 130 s awake at 0.264 W is 9.53 mWh, or 0.229 Wh per day at 24 readings [A1].
- **Uploads, sleep and valve.** Six uploads take 0.120 Wh, the sleep floor 4.0 mWh and the valve pulses 5.0 mWh per day [A2].
- **Daily need.** The sum is 0.358 Wh; with the 1.5 margin it is 0.537 Wh per day, a 22.4 mW average [A3]. The TRL 2 figure of about 0.5 Wh stands; the longer settling wait adds about 35 mWh per day [F3].
- **Sensitivity.** A site with only 2G coverage, where each upload costs four times as much, needs 1.077 Wh per day [A4]. The 15 min schedule allowed by R4 needs 1.589 Wh per day [A5].

■ B. Battery autonomy and solar recharge (R7)

- **Autonomy.** The 19.2 Wh pack gives 15.36 Wh usable, which lasts 28.6 days without sun, or 24.3 days at 0 °C. On a 2G-only site it lasts 14.3 days and on the 15 min schedule 9.7 days [B1]. R7 (7 days) is met in every case.
- **Recharge.** The 5 W panel yields 13.5 Wh on a clear day and 7.5 Wh on an overcast day [B2]. From empty it refills in 1.18 clear days or 2.21 overcast days [B3]. R7 (3 clear days) is met.

- **Panel size.** The smallest panel that would refill the pack in three overcast days is 3.8 W, so the 5 W panel carries a modest margin for the rainy season. At full sun it charges at about 1.25 A (0.21 C), well within the cells' rating [B4].
- **Hot weather.** Charging depends on the battery staying below its 45 °C charge limit (section C). Without the sun shield, a clear 45 °C day allows no useful charge at all.

■ C. Enclosure temperature (R9)

- **Loss coefficient.** The 200 x 120 x 260 mm enclosure has 0.2144 m² of surface, 0.1624 m² exposed, and loses 1.62 W/K to the air [C1].
- **Steady rise.** With the sun 60° high and square to the front face, a clean light grey enclosure absorbs 22.6 W and runs 13.9 K above ambient; a dusty one absorbs 35.2 W and runs 21.6 K above. The panel's shadow on the top reduces these to 8.7 K and 13.6 K [C2].
- **Time constant.** About 1,460 J/K of enclosure, cells and boards gives a 15 min time constant, so the inside follows the sun within the hour [C3].

Table 2. Hottest point inside the enclosure and charging on a clear day [C4].

Day	Case	Peak inside	Sun hours with charging allowed	Energy available
30 to 45 °C	Clean, no shield	60.7 °C	0.4 h	0.0 Wh
30 to 45 °C	Dusty, no shield	69.8 °C	0.2 h	0.0 Wh
30 to 45 °C	Clean, with shield	48.6 °C	6.3 h	7.3 Wh
30 to 45 °C	Dusty, with shield	50.8 °C	5.4 h	5.8 Wh
22 to 35 °C	Clean, no shield	50.8 °C	8.0 h	10.1 Wh
22 to 35 °C	Dusty, no shield	59.9 °C	0.4 h	0.0 Wh
22 to 35 °C	Dusty, with shield	40.9 °C	12.0 h	13.5 Wh

- **The TRL 2 enclosure would not charge in hot weather.** The battery's charge cut-off at 45 °C (R9) blocks charging for nearly all of a clear hot day, and even on a 35 °C day once the box is dusty. A run of such days would drain the pack within its 29 days.
- **Sun shield.** A ventilated white aluminum hood over the top, sides and front (BOM item 17, \$8) keeps the inside to about 51 °C on a 45 °C day and leaves 5.8 Wh for charging, 10.8 times the daily need. It lifts off for the monthly visit.
- **R9 is at risk.** The shield factor is assumed, the battery still sits above 45 °C for part of a hot day (which ages LiFePO₄ cells faster), and the electronics' ratings are met with margin (ESP32-S3 and SIM7000-class modules are rated to 85 °C). Enclosure sealing (IP66), UV-stable materials and the opaque cell are met by the choice of parts.

■ D. Sample flow, flushing and water use (R8, R1)

- **Cell volume.** The TRL 2 cell held 0.37 L. The TRL 3 cell holds 0.185 L of water once the baffle and the immersed sensors are subtracted. The 1/4 in sample tube from the tee to the cell runs 738 mm and holds 10.7 mL [D1].

- **Water use.** Each 90 s flush at 0.5 L/min uses 0.75 L, 18.0 L per day [D2]. At the start of the 10 s chlorine window, 2.9 % of the previous hour's water is still in the cell, and 1.8 % at its end; the TRL 2 cell would have held 17.0 %, enough to bias every chlorine reading low by that much when the old water had lost its residual [D3]. This is why the cell was made smaller rather than the flush longer.
- **Pressure.** A fixed orifice sized for 0.5 L/min at 1 bar passes 0.35 L/min at 0.5 bar and 1.00 L/min at 4 bar, which would use 36.0 L per day [D4] and break R8. A pressure-compensating regulator holds 0.5 L/min; at its +10 % tolerance the unit uses 19.8 L per day [D5]. **R8 is met on paper** with a 1 % margin at the upper tolerance. Below the regulator's compensating range, at 0.5 bar, the flow falls to about 0.35 L/min and the stale fraction rises to 8.5 % [D5]; a low-pressure site needs a longer flush, set per site.
- **Context.** The daily flush is 0.36 % of the draw of a tapstand serving 250 people at 20 L each. On the 15 min schedule the unit would use 72 L per day [D6]; R8 applies to hourly sampling.
- **Valve.** At 0.5 bar the valve must be direct acting, with no minimum pressure difference; pilot-operated valves need about 0.2 to 0.5 bar [D7]. BOM item 13 now says so.

■ E. Free chlorine error budget (R1)

The HOCl pKa is 7.75 at 5 °C, 7.54 at 25 °C and 7.43 at 40 °C [E1]. At 25 °C the share of free chlorine present as HOCl is 0.92 at pH 6.5, 0.77 at pH 7.0, 0.52 at pH 7.5, 0.26 at pH 8.0 and 0.10 at pH 8.5; across the pH and temperature range of R1 it spans 0.15 to 0.90 [E2]. The TRL 2 precis quoted about a quarter at pH 8 and three quarters at pH 7; both stand.

Calibration against a DPD comparator at the site absorbs the pH at the time of calibration into the sensor's slope. What remains is the change in pH between visits, the comparator's own reading error (± 0.07 mg/L), a 1.25 % residual temperature error and the 2.9 % stale-water bias [E3].

Table 3. Error in free chlorine at zero sensor drift, against the R1 target [E4].

Free chlorine	pH	Site pH (± 0.2 between visits)	pH probe variant (± 0.1)	Target
0.5 mg/L	7.0	± 0.092 mg/L	± 0.077 mg/L	± 0.10 mg/L
0.5 mg/L	7.5	± 0.131 mg/L	± 0.091 mg/L	± 0.10 mg/L
0.5 mg/L	8.0	± 0.202 mg/L	± 0.115 mg/L	± 0.10 mg/L
0.5 mg/L	8.5	± 0.260 mg/L	± 0.135 mg/L	± 0.10 mg/L
1.5 mg/L	7.0	± 0.194 mg/L	± 0.118 mg/L	± 0.225 mg/L
1.5 mg/L	7.5	± 0.339 mg/L	± 0.186 mg/L	± 0.225 mg/L
1.5 mg/L	8.0	± 0.574 mg/L	± 0.284 mg/L	± 0.225 mg/L
1.5 mg/L	8.5	± 0.753 mg/L	± 0.352 mg/L	± 0.225 mg/L

Bold values miss the target.

- **R1 is not met.** With a site pH, only 2 of the 8 cases meet the target, and those only at pH 7.0, with 0.038 mg/L left for sensor drift at 0.5 mg/L [E4], [E5]. A pH probe helps up to pH 7.5 but still misses at pH 8.0 and above. The sensor's own drift over a month is not in these figures, because no verified long-term data are available (see the gaps in docs/REVIEW.md).

- **The target is as tight as the field reference.** The DPD comparator alone takes ± 0.07 of the ± 0.10 mg/L budget. A target of ± 0.2 mg/L or ± 25 %, whichever is greater, would be met with a site pH up to pH 7.5 at zero drift [E6]. That change is proposed in `docs/REVIEW.md` , awaiting Amish; R1 is unchanged here.

■ F. Turbidity reading: bubbles and settling (R2)

- **Bubbles.** Air released when the flow stops scatters light like particles. At 5 °C a 100 μ m bubble rises at 3.6 mm/s and takes 18 s to clear the 66 mm cavity; a 50 μ m bubble takes 74 s. At 25 °C the same bubbles take 11 s and 43 s [F1]. The TRL 2 wait of 10 s was too short in cold water.
- **Settling.** Silt of 10 μ m settles at 0.10 mm/s: 3.5 mm during the 35 s of waiting and reading, and the clear layer at the top takes 5.4 min to reach the beam at mid-height [F2]. Settling does not bias the reading.
- **Change.** The wait is now 30 s, which clears 100 μ m bubbles at 5 °C, for 35 mWh per day [F3]. Smaller bubbles remain possible when the supply is supersaturated with air.
- **R2 is at risk.** Range and resolution follow the published open-source design (Kelley et al., 2014), but fouling of the optical windows between monthly cleans cannot be estimated on paper.

■ G. Data and storage (R6)

- **Airtime.** Each upload carries about 780 B of readings and status plus about 4,000 B of TLS, HTTP and network overhead, 0.86 MB per month; with TLS session resumption it falls to about 0.36 MB [G1]. R6's airtime assumption of under 1 MB per month holds.
- **Storage.** 90 days of readings take 138 kB as CSV on the microSD card; a 7-day ring buffer in flash takes 5.4 kB [G2]. R6 is met on paper.

■ H. Alert latency (R5)

- **From confirmation to SMS.** A modem wake, network attach and SMS take about 140 s at the first try; with three tries at 3 min spacing the worst case is 13.0 min [H1]. R5 (15 min) is met on paper where there is coverage. The TRL 2 plan did not fix the retry spacing; at 5 min spacing the worst case would be 17 min, so 3 min is now specified.
- **From the event.** Because two consecutive hourly readings confirm a crossing, the worst case from the start of an event to the SMS is 122 min. A confirming reading 15 min after the first crossing would cut this to 77 min [H2]; this is offered as a suggestion in `docs/REVIEW.md` , not adopted.

■ I. Pole and footing in wind (structure, R11)

- **Loads.** A 35 m/s gust (735 Pa) puts 42 N on the panel at 2.16 m, 50 N on the enclosure at 1.30 m and 89 N on the pole at 1.05 m: 181 N in all and a base moment of 249 N·m [I1]. The shield adds a little to the enclosure area; it is inside the margins below.
- **Pole.** The 48.3 x 3.2 mm tube has a section modulus of 4.80 cm³ and sees 52 MPa, a factor of 4.5 on the 235 MPa yield [I2].
- **Footing.** A 300 mm footing 600 mm deep in medium sand resists 886 N sideways, a factor of 4.9 [I3]. Soft or wet ground needs a site check.

■ J. Installation and the monthly visit (R11, R10)

- **Installation.** The tasks take 160 person-minutes. The critical path is digging, setting the pole, waiting 30 min for fast-set concrete, then mounting, cabling and commissioning: 140 min against the 120 min of R11 [J1]. With the footing cast on an earlier visit, the critical path on the day is 65 min [J2]. **R11 is at risk** on the 2 h target; the design parts of R11 (one tee, an isolation valve, a check valve, food-safe wetted parts and hand tools only) are met by design, and the open air break at the outlet keeps the drain from ever connecting back to the cell.
- **Monthly visit.** Cleaning the cell and electrode, a DPD test, a pH check, calibration from a phone and a look at the strainer, drain and panel take 30 min, exactly the R10 limit [J3]. **R10 is not met on present evidence:** a monthly visit is only enough if the chlorine sensor drifts less than about 0.04 mg/L in a month (section E), which no verified data yet show.

■ K. Cost (R12)

The BOM has 17 lines totaling \$282.00 against the \$300 budget_usd , a margin of \$18.00 (6 %) [K1]. The TRL 3 changes added \$3 for the pressure-compensating regulator, \$1 for the air-break fitting and \$8 for the sun shield. R12 is met on paper. A pH probe variant (about \$60 to \$100, estimate) would not fit.

■ L. Results against every requirement

Table 4. Requirement status from this note [L].

ID	Requirement	Value	Target	Status
R1	Measure free chlorine	±0.131 mg/L at 0.5 mg/L and pH 7.5 with site pH, before drift; 2 of 8 cases met	±0.1 mg/L below 1.0 mg/L, ±15 % above, pH 6.5 to 8.5	Not met
R10	Maintainable by a caretaker	Visit 30 min; needs drift below about 0.04 mg/L per month, unknown	Monthly or less, 30 min or less	Not met on present evidence
R2	Measure turbidity	30 s wait clears 100 µm bubbles at 5 °C; fouling unknown	0 to 100 NTU, ±0.5 NTU or ±10 %	At risk
R9	Survive outdoors	50.8 °C peak inside on a 45 °C day with the shield (69.8 °C without)	IP65; 0 to 45 °C ambient; charging blocked above 45 °C	At risk (shield factor assumed)
R11	Install simply and safely	140 min critical path; 65 min with the footing cast earlier	2 h, hand tools, backflow prevented	At risk
R5	Alert the people who act	13.0 min worst case from confirmation to SMS	15 min	Met on paper (where there is coverage)
R6	Report and keep data	0.86 MB per month; 90 days in 138 kB	4 h uploads, 90 days on board	Met on paper
R7	Run on sunlight alone	28.6 days autonomy; 1.18 clear days to refill	7 days; 3 clear days	Met on paper
R8	Use little treated water	18.0 L/day nominal, 19.8 L at regulator tolerance	20 L/day	Met on paper (thin margin)
R12	Stay within the budget	\$282	\$300	Met on paper
R3	Measure water temperature	DS18B20, ±0.5 °C from -10 to 85 °C	0 to 50 °C, ±0.5 °C	Met by design
R4	Sample on a schedule	Hourly default; 15 min schedule within energy (1.589 Wh/day, 9.7 days)	15 min to 24 h	Met by design
R13	Report measurements, not verdicts	Alert wording rule	Never says "safe"	Met by design

Counts: 2 not met, 3 at risk, 5 met on paper, 3 met by design. No requirement is left unverifiable at TRL 3, although R2, R9 and R10 each rest on an assumption that only a test can settle.

■ Checks against the TRL 2 figures

TRL 2 claim (WWT-PRC-001 v0.2)	This note	Action
About 0.5 Wh/day	0.537 Wh/day (130 s awake)	Precis updated
About 30 days without sun	28.6 days (24.3 days at 0 °C)	Precis updated
About 13 Wh per clear day; about 1.2 days to refill	13.5 Wh; 1.18 days	Stands
Cell 0.37 L; 0.75 L flush is about two cell volumes	0.185 L; 0.75 L is about four cell volumes	Cell halved, precis updated
About 18 L/day, R8 met	18.0 L nominal, but 36 L at 4 bar with a fixed restrictor	Regulator added, precis updated
10 s wait for bubbles	Too short at 5 °C; 30 s	Precis updated
Enclosure temperature not estimated	61 to 70 °C peak without a shield on a 45 °C day	Shield added, precis updated
Under 1 MB per month	0.86 MB	Stands
SMS within minutes	13.0 min worst case with 3 min retries	Precis updated
HOCl about a quarter at pH 8, three quarters at pH 7	0.26 and 0.77	Stands
About \$270	\$282	Precis and BOM notes updated